

SYED MUHAMMAD RAFAY ALI

Full Stack Developer

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SUMMARY

Computer Science graduate from FAST-NUCES with hands-on experience building and shipping full-stack web applications using React, TypeScript, Node.js, and Express. Designed and deployed RESTful APIs, integrated third-party services, and set up CI/CD pipelines with GitHub Actions and Docker across five end-to-end projects spanning booking platforms, AI-driven event management, and interactive 3D interfaces. Comfortable across the full stack, from responsive cross-browser frontend interfaces to PostgreSQL and MongoDB backends, with additional exposure to machine learning techniques including neural networks and K-Means clustering.

EDUCATION

Bachelor of Science in Computer Science

2022 – 2026

National University of Computer and Emerging Sciences (FAST-NUCES), Lahore

SKILLS

Languages: C/C++, JavaScript, TypeScript, Python

Frontend: React, React Router, HTML5, CSS3, Tailwind CSS, Bootstrap, jQuery, Vite, responsive design, cross-browser compatibility

Backend: Node.js, Express.js, RESTful APIs, JWT, Nodemon

Databases & DevOps: MongoDB, PostgreSQL, Supabase, Docker, GitHub Actions, CI/CD, Git, Vercel, Railway

Spoken Languages: English (Fluent), Urdu (Fluent)

PROJECTS

HostelSpot: Student Hostel Booking Platform | *React, TypeScript, Supabase, Tailwind CSS*

- Built a full-stack hostel booking platform with dual dashboards for students and hostel owners, real-time availability updates, and an admin panel; designed a responsive, cross-browser-compatible UI using Tailwind CSS.
- Developed RESTful API endpoints integrated with Supabase for authentication and real-time data sync, reducing average booking confirmation time by ~60% compared to manual processes.
- Configured a CI/CD pipeline using GitHub Actions and Docker to automate testing and deployment.

FestivAI: AI-Powered Event Management Platform | *React, Node.js, Express, MongoDB*

- Engineered a full-stack event-management platform with RSVP-driven booking, dynamic QR check-ins, venue management, and AI-generated e-card invitations.
- Integrated Google Maps API for real-time hall location discovery and built RESTful API endpoints in Express/Node.js connecting a React frontend to a MongoDB backend.
- Implemented JWT-based authentication and role-based access control for secure separation between organizer and attendee permissions.

CarInsight Pro: Car Research & Comparison Platform | *React, Node.js, MongoDB, Express, Railway*

- Built a MERN-stack platform for side-by-side car specification comparisons with an interactive matchmaking quiz and a custom CMS; responsive UI compatible across major browsers.
- Designed RESTful API routes in Express to serve structured car data from MongoDB, supporting filtering, sorting, and comparison queries.
- Configured a CI/CD pipeline using GitHub Actions, running 7 Jest and Supertest integration tests against an in-memory MongoDB instance on every push before deployment to Railway.

PakTravel AI: Intercity Bus Travel System | *Python, scikit-learn, Pandas, NumPy*

- Built an AI-powered travel system combining classical search algorithms, propositional logic, and CSP-based scheduling to model real-world intercity bus route constraints.
- Applied neural networks and K-Means clustering for route optimisation and demand analysis across intercity travel patterns.
- Processed and cleaned structured travel datasets using Pandas and NumPy, enabling accurate model training and result validation across multiple route configurations.

Interactive 3D Portfolio | *React, TypeScript, Vite, Tailwind CSS*

- Built a visually rich personal portfolio featuring a custom mathematical SVG character with cursor-tracking eyes, custom shaders, and scroll-based reveal animations, all without third-party animation libraries.
- Engineered smooth performance across devices by optimising render cycles and implementing lazy-loaded sections, demonstrating strong frontend architecture and attention to user experience.